

(1001CJA102120108)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
03-07-2021
JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	D	B	A	B	B	B	A	A
SECTION-II	Q.	11	12	13	14	15	16				
	A.	C	B	B,C,D	A,B,C,D	A	B				
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		P,R	Q,T	R	S,T		P,R	P,Q,R,S,T	P,R	P,R	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	C	B	C	D	C	B	D	A,B,D
SECTION-II	Q.	11	12	13	14	15	16				
	A.	C	A,B,D	A,C,D	C,D	B,C,D	D				
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		Q,S,T	P,Q,S,T	R,T	P,Q,S,T		P,Q,S	T	Q,S	R	

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	D	A	B	B	B	A	D	C
SECTION-II	Q.	11	12	13	14	15	16				
	A.	B,D	A,B,C,D	B	A	C	B,D				
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		Q	Q,S	Q,S	S		P	Q	R	S	

(1001CJA102120109)

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JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C,D	B,C	A,C,D	A,B	A,B,C,D	C,D	A,B	A,B,D	C	B
	Q.	11	12								
SECTION-II	A.	C	D								
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	15.00	4.84	20.00	1.25	1.50	16.36	0.10	4.68	1.70 to 1.75	29.60

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C	A,B,D	A,B,C,D	A,B,D	B,D	A,B,C,D	A,B,C,D	A,C,D	A	B
	Q.	11	12								
SECTION-II	A.	B	D								
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	18.33	20.00	4.00	5.00	6.00	3183.00	2.00	3.00	4.00	7.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,C	B	A,B,C	A,B,C	A,C,D	D	A,B,C	B	A
	Q.	11	12								
SECTION-II	A.	D	A								
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.00	2.00	2.70	10.00	0.80	638.00	7.00	7.40	2.00	0.00



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JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

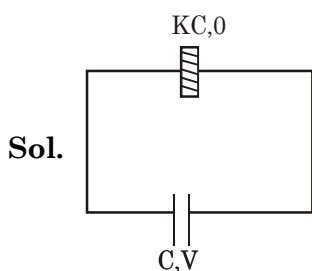
PAPER-1

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (D)



$$V' = \frac{(KC)(0) + (C)(V)}{KC + C}$$

$$V' = \frac{CV}{KC + C}$$

$$KV' + V' = V$$

$$KV' = V - V'$$

$$K = \frac{V - V'}{V'}$$

2. Ans. (A)

Sol. $\phi = at(T - t) = aTt - at^2$

$$\varepsilon = -\frac{d\phi}{dt} = -(aT - 2at)$$

$$P = -\frac{\varepsilon^2}{R} = \frac{(aT - 2at)^2}{R}$$

$$H = \int_0^T P dt = \int_0^T \frac{(aT - 2at)^2}{R} dt$$

$$= \frac{a^2 T^3}{3R}$$

3. Ans. (D)

Sol. Current in AC circuit,

$$I = \frac{V_0}{X_L - X_C} \sin(\omega t - \frac{\pi}{2}) = \frac{V_0 \omega C}{1 - \omega^2 LC} \cos \omega t$$

then

$$V_C = \frac{1}{\omega C} \left(\frac{V_0 \omega C}{1 - \omega^2 LC} \right) \cos(\omega t - \frac{\pi}{2}) = \frac{V_0}{1 - \omega^2 LC} \sin \omega t$$

charge on capacitor

$$Q = CV_C = \left(\frac{V_0 C}{1 - \omega^2 LC} \right) \sin \omega t$$

Now let at time t key is shifted to position 2, then for maximum charge on capacitor at later time.

$$\frac{1}{2} \frac{Q_{\max}^2}{C} = \frac{1}{2} \frac{Q^2}{C} + \frac{1}{2} LI^2$$

$$= \frac{1}{2C} \left(\frac{V_0 C}{1 - \omega^2 LC} \sin \omega t \right)^2 + \frac{1}{2} L \left(\frac{V_0 \omega C}{1 - \omega^2 LC} \cos \omega t \right)^2$$

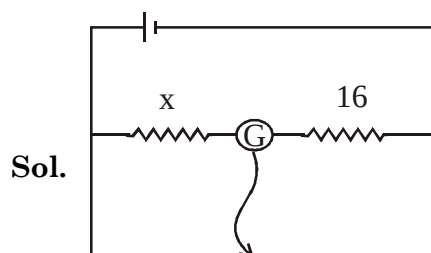
$$= \frac{V_0^2 C}{2(1 - \omega^2 LC)^2} \sin^2 \omega t + \frac{V_0^2 \omega^2 C^2 L}{2(1 - \omega^2 LC)^2} \cos^2 \omega t$$

$$= \frac{V_0^2 C}{2(1 - \omega^2 LC)^2} [\sin^2 \omega t + \omega^2 LC \cos^2 \omega t]$$

4. Ans. (B)

Sol. As is evident from the graph $v_1 < v_2 < v_3$, V_{10} , V_{20} , V_{30} in this case are the stopping potential.

5. Ans. (A)



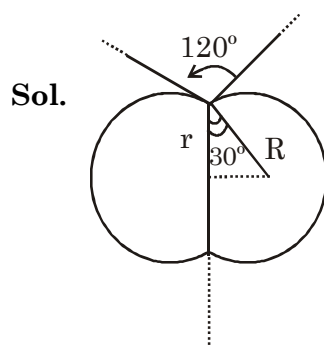
$$\frac{x}{16} = \frac{36}{64} \Rightarrow x = \frac{36 \times 16}{64} = 9\Omega$$

$$\frac{dx}{x} = \frac{d\ell \times 100}{\ell(1-\ell)} = \frac{0.1 \times 100}{36 \times 64}$$

$$\Rightarrow dx = \frac{9 \times 10}{36 \times 64} = \frac{5}{128} \Omega$$

$$x = \left(9 \pm \frac{5}{128} \right) \Omega$$

6. Ans. (B)



Because of surface tension the angle between tangents shown = 120° also the common interface is flat.

The radius of common interface = $R \cos 30^\circ$

$$\text{So area} = \pi r^2 = \pi \left(\frac{\sqrt{3}R}{2} \right)^2$$

$$= \frac{3\pi R^2}{4}$$

$$R \cos 30^\circ = \frac{\sqrt{3}R}{2}$$

$$\pi \cdot \left(\frac{\sqrt{3}R}{2} \right)^2$$

$$\frac{3\pi R^2}{4}$$

7. Ans. (B)

Sol. Let l is natural length and K is spring force constant

$$T_1 = K(a_1 - l) \quad \dots\dots(i)$$

$$T_2 = K(a_2 - l) \quad \dots\dots(ii)$$

From (i) and (ii)

$$K = \frac{T_1 - T_2}{a_1 - a_2}$$

$$l = \frac{T_1 a_2 - T_2 a_1}{T_1 - T_2}$$

The amount of work done

$$W = \frac{1}{2} K(a_1 + a_2 - l)^2$$

8. Ans. (B)

Sol. For $R = 4r$, the sequence repeats itself.

9. Ans. (A)

10. Ans. (A)

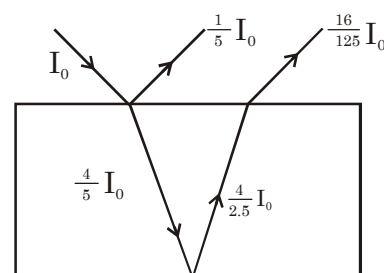
11. Ans. (C)

Sol. Light undergoes inversion at both interface, so for destructive interference

$$2\mu d = \frac{\lambda}{2}$$

$$d = \frac{\lambda}{4\mu} = \frac{\lambda}{4.8}$$

12. Ans. (B)



$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{\max} = \left(\sqrt{\frac{I_0}{5}} + \sqrt{\frac{16}{125} I_0} \right)^2 = \frac{81}{125} I_0$$

13. Ans. (B,C,D)

Sol. Frictional force and its torque are equal and opposite on A and B.

$$\text{If } \alpha_B = \alpha \text{ (+ve)}$$

$$\text{then } \alpha_A = -\alpha \text{ (-ve)}$$

$$\text{Total torque } \tau = 0$$

$$\therefore I\omega_0 = I\omega_A + I\omega_B$$

$$\text{or } \omega_A + \omega_B = \omega_0 \quad \dots (1)$$

$$\omega_A = \omega_0 - \alpha t \text{ for } t < t_0$$

$$\omega_B = \alpha t \text{ for } t < t_0$$

$$\omega_A = \omega_B = \frac{\omega_0}{2} \text{ for } t > t_0$$

$$\text{i.e. } \omega_A - \omega_B = \omega_0 - 2\alpha t \quad \dots (2)$$

$$\alpha_A + \alpha_B = 0 \quad \dots (3)$$

$$\text{and } \alpha_A - \alpha_B = -2\alpha \quad \dots (4)$$

$$\alpha_B - \alpha_A = +2\alpha \quad \dots (5)$$

14. Ans. (A,B,C,D)

15. Ans. (A)

16. Ans. (B)

Sol. $\lambda_m \cdot T = b$

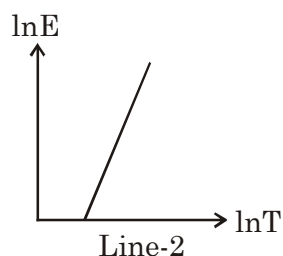
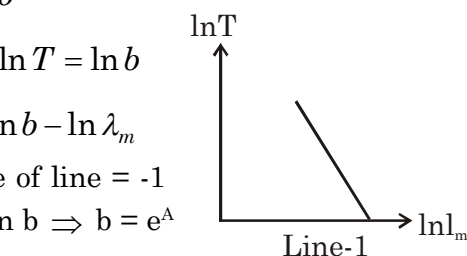
$$\ln \lambda_m + \ln T = \ln b$$

$$\ln T = \ln b - \ln \lambda_m$$

$$\Rightarrow \text{Slope of line} = -1$$

$$\Rightarrow A = \ln b \Rightarrow b = e^A$$

$$\text{and } E = \sigma \cdot T^4 \cdot A$$



$$\ln E = \ln (\sigma A) + 4 \ln T$$

$$\Rightarrow \text{slope} = +4$$

$$\Rightarrow \ln T = \ln b - \ln \lambda_m$$

$$\ln T = A - C$$

SECTION-II

1. Ans. 2

Sol. Let C be the capacitance of the capacitor when the pressure is P and C be the capacitance when the pressure is P + ΔP. We'll assume that (a) the change in the thickness of the plates is small, and (b) the total volume of material between the plates is conserved. We can use the expression for the capacitance of a dielectric-filled parallel-plate capacitor and the definition of Young's modulus to express the change in the capacitance ΔC of the given capacitor when the pressure on its plates is increased by ΔP.

Express the change in capacitance resulting from the decrease in separation of the capacitor plates by Δd:

$$\Delta C = C' - C = \frac{\kappa \epsilon_0 A'}{d - \Delta d} - \frac{\kappa \epsilon_0 A}{d}$$

Because the volume is constant:

$$A'd' = Ad$$

or

$$A' = \left(\frac{d}{d'} \right) A = \left(\frac{d}{d - \Delta d} \right) A$$

Substitute for A' in the expression for ΔC and simplify to obtain:

$$\Delta C = \frac{\kappa \epsilon_0 A}{d - \Delta d} \left(\frac{d}{d - \Delta d} \right) - \frac{\kappa \epsilon_0 A}{d}$$

$$= \frac{\kappa \epsilon_0 A}{d(d - \Delta d)^2} d^2 - \frac{\kappa \epsilon_0 A}{d}$$

$$= \frac{\kappa \epsilon_0 A}{d} \left[\frac{d^2}{(d - \Delta d)^2} - 1 \right]$$

$$= C \left[\frac{d^2}{(d - \Delta d)^2} - 1 \right]$$

From the definition of Young's modulus :

$$\frac{\Delta d}{d} = -\frac{P}{Y} \Rightarrow \Delta d = -\left(\frac{P}{Y} \right) d$$

Substitute for Δd in the expression of ΔC to obtain :

$$\Delta C = \frac{\kappa \epsilon_0 A}{d} \left[\frac{d^2}{\left\{ d + \left(\frac{P}{Y} \right) d \right\}^2} - 1 \right]$$

$$= C \left[\left\{ 1 + \left(\frac{P}{Y} \right) \right\}^{-2} - 1 \right]$$

Expand $\left(1 - \frac{P}{Y} \right)^{-2}$ binomially to obtain :

$$\left(1 - \frac{P}{Y} \right)^{-2} = 1 - 2 \frac{P}{Y} + 3 \left(\frac{P}{Y} \right)^2 + \dots$$

Provided $P \ll Y$:

$$\left(1 - \frac{P}{Y} \right)^{-2} \approx 1 - 2 \frac{P}{Y}$$

Substitute in the expression for ΔC and simplify to obtain :

$$\Delta C = C \left[1 - 2 \frac{P}{Y} - 1 \right] = \boxed{-2 \frac{P}{Y} C}$$

2. Ans. 1.00

Sol. $t = \frac{dE [B^2 \ell^2 C + M]}{B \ell m g} = 1000 \text{ sec} = 1$

3. Ans. 544.90 to 545.00

Sol. The buoyant force on the balloon must be equal to the weight of the load plus the weight of the gas.

The buoyant force is $F_B = \rho_{\text{air}} V g$. A lift of

290 kg means $\frac{F_B}{g} - m_{\text{hot}} = 290 \text{ kg}$, where

m_{hot} is the mass of hot air in the balloon.
 $m = \rho V$.

$$m_{\text{hot}} = \rho_{\text{hot}} V \cdot \frac{F_B}{g} - m_{\text{hot}} = 290 \text{ kg gives}$$

$$(\rho_{\text{air}} - \rho_{\text{hot}}) V = 290 \text{ kg}$$

Solving for ρ_{hot} gives

$$\rho_{\text{hot}} = \rho_{\text{air}} - \frac{290 \text{ kg}}{V} = 1.23 \text{ kg/m}^3$$

$$-\frac{290 \text{ kg}}{500.0 \text{ m}^3} = 0.65 \text{ kg/m}^3 \cdot \rho_{\text{hot}} = \frac{pM}{RT_{\text{hot}}}$$

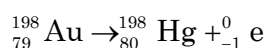
$$\rho_{\text{air}} = \frac{pM}{RT_{\text{air}}} \cdot \rho_{\text{hot}} T_{\text{hot}} = \rho_{\text{air}} T_{\text{air}} \text{ so}$$

$$T_{\text{hot}} = T_{\text{air}} \left(\frac{\rho_{\text{air}}}{\rho_{\text{hot}}} \right) = (288 \text{ K})$$

$$\left(\frac{1.23 \text{ kg/m}^3}{0.65 \text{ kg/m}^3} \right) = 545 \text{ K} = 272^\circ \text{C}$$

4. Ans. 0.96

Sol. Find the energy equivalent of the mass decrease. Part of the released energy appears as the emitted photon and the rest as kinetic energy of the electron.



The mass change is $197.968225 \text{ u} - 197.966752 \text{ u} = 1.473 \times 10^{-3} \text{ u}$

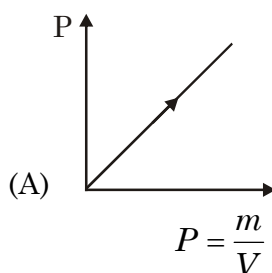
(The neutral atom masses include 79 electrons before the decay and 80 electrons after the decay. This one additional electron in the product accounts correctly for the electron emitted by the nucleus.)

The total energy released in the decay is $(1.473 \times 10^{-3} \text{ u}) (931.5 \text{ MeV/u}) = 1.372 \text{ MeV}$. This energy is divided between the energy of the emitted photon and the kinetic energy of the β^- particle. Thus the β^- particle has kinetic energy equal to $1.372 \text{ MeV} - 0.412 \text{ MeV} = 0.960 \text{ MeV}$.

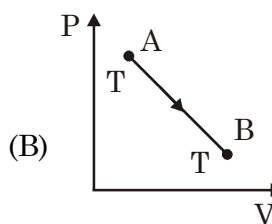
SECTION-IV

1. Ans. (A) $A \rightarrow P, R$; $B \rightarrow Q, T$; $C \rightarrow R$; $D \rightarrow S, T$

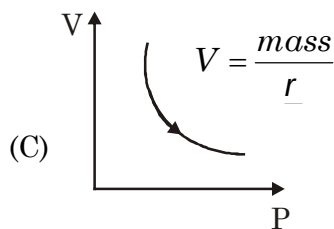
Sol. List - I List - II



(P) Temp. constant Int. energy constant work done negative heat released by the gas.

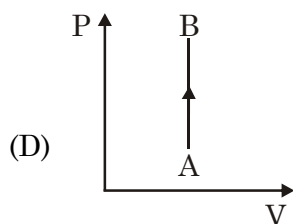


(Q) Work done is positive heat absorbed by the gas.



(R) Process is not clear but volume is decreasing.

$\Rightarrow$ Work done is negative.



(S) Work done is zero. Int energy increases. Heat is absorbed by the gas.

2. **Ans. (A)-(P,R); (B)-(P,Q,R,S,T); (C)-(P,R); (D)-(P,R)**

Sol. For $P = \frac{1}{f} = (1.5 - 1) \left(\frac{1}{+6} - \frac{1}{+15} \right) = \frac{1}{2} \left(\frac{15 - 6}{90} \right)$

$$= \frac{1}{2} \times \frac{9}{90} = \frac{1}{20} \Rightarrow f = +20 \text{ cm}$$

$$\text{Power} = \frac{1}{+20} > 0 \Rightarrow D \text{ matching}$$

$$\text{For } u = -20, \quad \frac{1}{v} - \frac{1}{-20} = \frac{1}{+20} \Rightarrow v \rightarrow \infty \Rightarrow A \text{ is matching}$$

$$\text{For } u = -40 \text{ cm}, \quad v = +40 \text{ cm} \Rightarrow B \text{ is matching}$$

For $u = -10$

$$\frac{1}{v} - \frac{1}{-10} = \frac{1}{20} \Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{10} = -\frac{1}{20}$$

$$v = -20 \text{ cm} \Rightarrow C \text{ is matching}$$

For

$$q = \frac{1}{f} = (1.5 - 1) \left(\frac{1}{-2} - \frac{1}{-2.5} \right) = \frac{1}{2} \left(\frac{-1}{2} + \frac{2}{5} \right)$$

$$= \frac{-5 + 4}{20} = \frac{-1}{20}$$

$$f = -20 \text{ cm}$$

$$\text{Power} = \frac{1}{f} < 0, \quad D \text{ is not matching}$$

For $u = -20 \text{ cm}$,

$$\frac{1}{v} - \frac{1}{-20} = \frac{1}{-20} \Rightarrow \frac{1}{v} = \frac{-1}{20} - \frac{1}{20} = \frac{-1}{10}$$

$$v = -10 \text{ cm} \Rightarrow A \text{ is not matching}$$

For $u = -10 \text{ cm}$

$$\Rightarrow \frac{1}{v} - \frac{1}{-10} = \frac{1}{-20} \Rightarrow \frac{1}{v} = \frac{-1}{20} - \frac{1}{10} = \frac{-3}{20}$$

$$v = -\frac{20}{3} \text{ cm} \Rightarrow C \text{ is not matching}$$

For $u = +40 \text{ cm}$,

$$\frac{1}{v} - \frac{1}{+40} = \frac{1}{-20} \Rightarrow \frac{1}{v} = \frac{-1}{20} + \frac{1}{40} = \frac{-2 + 1}{40} = \frac{-1}{40}$$

$$\Rightarrow v = -40 \text{ cm} \Rightarrow B \text{ is matching}$$

$$\text{For } r, \Rightarrow \left(\frac{-1}{f} \right) = \left(\frac{\frac{4}{3} - 1}{+2w/3} \right) \times 2 + \left(\frac{-2 \times \frac{4}{3}}{-2w/3} \right)$$

$$= \frac{1}{200} \times 2 + \frac{4}{100} = \frac{5}{100} = \frac{1}{20}$$

$f = -20 \text{ cm} \Rightarrow$ concave mirror power is +ve
if $u = -20 \text{ cm} \Rightarrow$ focus its image will be at infinity

if $u = -40 \text{ cm} = 2f$ image will also be at -40 cm

if $u = -10 \text{ cm}$ by mirror formula,

$$\frac{1}{v} + \frac{1}{-10} = \frac{1}{-20} \Rightarrow \frac{1}{v} = \frac{-1}{20} + \frac{1}{10} = \frac{1}{20} \Rightarrow v = +20 \text{ cm}$$

for s, $f = +20 \text{ cm}$, power is -ve

for $u = -20 \text{ cm}$,

$$\frac{1}{v} + \frac{1}{-20} = \frac{1}{+20} \Rightarrow v = +10 \text{ cm}$$

A not correct

for $u = +40 \text{ cm}$

$$\Rightarrow \frac{1}{v} + \frac{1}{40} = \frac{1}{+20} \Rightarrow v = +40 \text{ cm},$$

B is correct.

$$\text{for } u = -10, \quad \frac{1}{v} + \frac{1}{-10} = \frac{1}{20} \Rightarrow v = \frac{+20}{3} \text{ cm},$$

C is not correct

$$\text{for } t \quad -\frac{1}{f} = 2 \left(\frac{\frac{4}{3} - 1}{-\frac{40}{3}} \right) + \left(\frac{-2 \times \frac{4}{3}}{\infty} \right);$$

$$-\frac{1}{f} = 2 \frac{\frac{4}{3} - 1}{-\frac{40}{3}}$$

$f = +20 \text{ cm}$ like convex mirror

$\Rightarrow S \& T$ are similar.

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**1. **Ans. (D)**

$$\Delta T_b = 100.975 - 100 = 0.975^\circ\text{C}$$

$$\text{Molecular weight of } \text{Ca}(\text{NO}_3)_2 = 164$$

$$\Delta T_b = i \times K_b \times \text{molality}$$

$$0.975 = i \times 0.52 \times \frac{1.23/164}{10/1000}$$

$$i = 2.5$$

$$\text{Degree of ionisation, } \alpha = \frac{i-1}{n-1} = \frac{2.5-1}{3-1}$$

$$= 0.75 \text{ (Ans.)}$$

2. **Ans. (D)**3. **Ans. (C)**

Based on electronic configuration.

4. **Ans. (B)**

Let us assume that the reaction starts at 10 minutes i.e. $t = 20 - 10 = 10$ minutes.

$$k = \frac{2.303}{t} \log \left(\frac{[A]_0}{[A]_t} \text{ or } \frac{r_0}{r_t} \right) \quad (r \rightarrow \text{rate of}$$

reaction)

$$k = \frac{2.303}{10} \cdot \log \frac{0.4}{0.04} \Rightarrow k = \frac{2.303}{10}$$

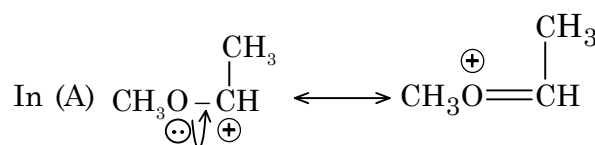
$$\frac{t_1}{2} = \frac{0.693}{k} \Rightarrow t_{1/2} = 3 \text{ minutes}$$

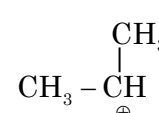
5. **Ans. (C)**

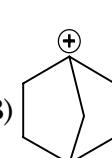
$\text{F}-\text{CH}_2-\text{CH}_2-\text{F}$, gauche form has greater stability due to gauche effect, then antiform.

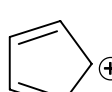
6. **Ans. (D)** Keesom forces are present between two polar covalent molecules.7. **Ans. (C)** $\Delta H_{\text{ion}}^\circ = 13.7 - 11 = 2.7 \text{ kcal/eq.}$
n.f. of H_3PO_4 against $\text{KOH} = 3$

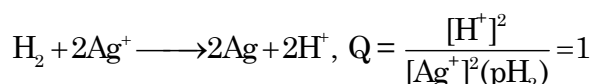
$$\Delta H_{\text{ion}}^\circ = 2.7 \times 3 \text{ kcal/mole} = 8.1 \text{ kcal/mole.}$$

8. **Ans. (B)** Rate of $\text{S}_{\text{N}}1$ depends upon stability of carbocation formed after removal of leaving group.

is more stable than 

In (B)  Highly unstable (Bredt's rule)

in option (C)  $\oplus$ is Aromatic  $\oplus$ is Anti aromatic

9. **Ans. (D)** Initial condition

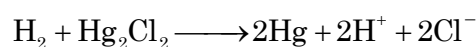
$$E_1 = E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.06}{2} \log(1) = E_{\text{cell}}^\circ \quad \dots (i)$$

After adding NaOH , pH of L.H.S. compartment become 7

$$Q = \frac{(10^{-7})^2}{(0.1)^2 \times 1} = 10^{-12}$$

$$E_2 = E_{\text{cell}}^\circ - \frac{0.06}{2} \log 10^{-12} = (E_{\text{cell}}^\circ + 0.36) \quad \dots (ii)$$

$$E_2 - E_1 = 0.36 \text{ V}$$

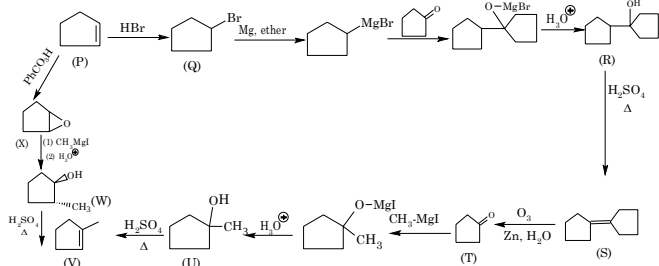
10. **Ans. (A, B, D)**

$$Q = \frac{[\text{H}^+]^2 [\text{Cl}^-]^2}{P_{\text{H}_2}}$$

$$E_{\text{cell}} = 0.28 - \frac{0.06}{2} \log \frac{[\text{H}^+]^2 [\text{Cl}^-]^2}{P_{\text{H}_2}}$$

11. Ans. (C)
12. Ans. (A,B,D)
13. Ans. (A,C,D)
14. Ans. (C, D)

Sol.



15. Ans. (B,C,D)

$$\Delta G = nRT \ln \left(\frac{P_2}{P_1} \right) = 12 \ln \left(\frac{2}{6} \right) = -12 \times 1.1 = -13.2$$

litre-atm

$$\Delta H = 0 \text{ and } \Delta U = 0 \quad (\Delta T = 0)$$

16. Ans. (D)

SECTION-II

1. Ans. (50.00)
2. Ans. (4.00)
3. Ans. (5.00)
4. Ans. (5.00)

SECTION-IV

1. Ans. (A)-(Q,S,T); (B)-(P,Q,S,T); (C)-(R,T); (D)-(P,Q,S,T)
2. Ans. (A)-(P,Q,S); (B)-(T); (C)-(Q,S); (D)-(R)

PART-3 : MATHEMATICS

SECTION-I

1. Ans. (C)

Sol. Since $f(x_1) = f(x_2) \quad \forall x_1, x_2 \in \mathbb{R}$

$\therefore f(x)$ must be constant

$$\text{Hence } p^2 - 1 = 0 \Rightarrow p = \pm 1$$

$$q^2 + 2q - 3 = 0 \Rightarrow q = -3, 1$$

$\therefore$ Largest possible value of $|p + q| = 4$

2. Ans. (B)

Sol. $x^n - 1 = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3) \dots (x - \alpha_n)$

By Putting $x = 2$,

$$2^n - 1 = (2 - \alpha_1)(2 - \alpha_2) \dots (2 - \alpha_{n-1})$$

$$\Rightarrow |2 - \alpha_1| |2 - \alpha_2| \dots |2 - \alpha_{n-1}|$$

$$= |2^n - 1| = 2^n - 1.$$

3. Ans. (D)

Sol. $(1+x)^{101} = ({}^{101}C_0 + {}^{101}C_1x + {}^{101}C_2x^2 + \dots)$

$$(1+x)^{100} = ({}^{100}C_0 + {}^{100}C_1x + \dots + {}^{100}C_{100}x^{100})$$

$$(1+x)^{201} = ({}^{101}C_0 + {}^{101}C_1x + \dots)$$

$$({}^{100}C_{100}x^{100} + {}^{100}C_{99}x^{99} + \dots)$$

SOLUTION

$\therefore$ coefficient of x^{100} is

$${}^{201}C_{100} = {}^{101}C_0 {}^{100}C_{100} + {}^{101}C_1 {}^{100}C_{99} + \dots$$

$$\text{Also } (1-x)^{100}$$

$$= ({}^{100}C_0 - {}^{100}C_1x + {}^{100}C_2x^2 - \dots - {}^{100}C_{99}x^{99} + {}^{100}C_{100}x^{100})$$

$$\therefore (1+x)^{101} (1-x)^{100} = ({}^{101}C_0 + {}^{101}C_1x + {}^{101}C_2x^2 + \dots)$$

$$({}^{100}C_{100}x^{100} - {}^{100}C_{99}x^{99} + \dots)$$

$\therefore$ coeff. of x^{100} of

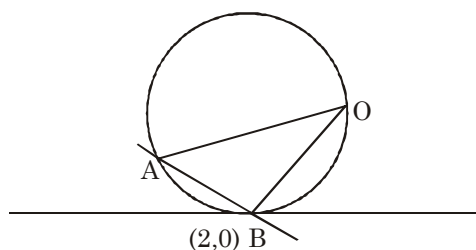
$${}^{100}C_{50} = {}^{101}C_0 {}^{100}C_{100} - {}^{101}C_1 {}^{100}C_{99} + {}^{101}C_2 {}^{100}C_{98} - \dots$$

Adding the two

$$\Rightarrow {}^{201}C_{100} + {}^{100}C_{50} = 2 [{}^{101}C_0 {}^{100}C_{100} + {}^{101}C_2 {}^{100}C_{98} + \dots]$$

4. Ans. (A)

Sol. Circle touches the x-axis at point B.



Slope of line AB, $|M|$ = Angle between OA & OB

$$|M| = \left| \frac{\frac{1}{2} - \frac{1}{7}}{1 + \frac{1}{2} \times \frac{1}{7}} \right| = \frac{1}{3}$$

5. Ans. (B)

Sol. Let $\frac{1}{8} = \int_0^{\pi/2} \frac{x^2 (\cos x - \sin x) dx}{(\cos x + \sin x)^3}$

$$\Rightarrow \frac{1}{8} = \int_0^{\pi/2} \frac{x^2 \cos 2x dx}{(1 + \sin 2x)^2}$$

Let $2x = t$

$$\Rightarrow I = \int_0^{\pi} \frac{t^2 \cos t dt}{(1 + \sin t)^2}$$

By parts

$$= \left(\frac{-t^2}{1 + \sin t} \right)_0^{\pi} + 2 \int_0^{\pi} \frac{t}{1 + \sin t}$$

$$I = -\pi^2 + 2I_1$$

where $I_1 = \int_0^{\pi} \frac{t}{1 + \sin t} \dots(i)$

From king property

$$I_1 = \int_0^{\pi} \frac{\pi - t}{1 + \sin t} \dots(ii)$$

(i) + (ii)

$$\Rightarrow 2I_1 = \pi \int_0^{\pi} \frac{dt}{1 + \sin t}$$

$$\Rightarrow 2I_1 = \pi \int_0^{\pi} (\sec^2 t - \sec t - \tan t) dt$$

$$\Rightarrow \frac{2I_1}{\pi} = (\tan t - \sec t)_0^{\pi} \Rightarrow \frac{2I_1}{\pi} = 1 + 1$$

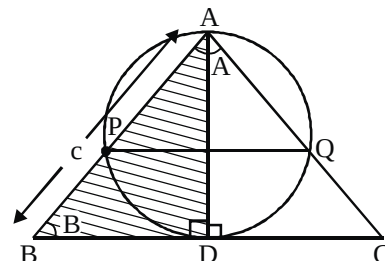
$$\Rightarrow I_1 = \pi \Rightarrow I = \pi(2 - \pi)$$

6. Ans. (B)

Sol. In $\triangle ABD$, $AD = c \sin B$

Now, in $\triangle APQ$, $\frac{PQ}{\sin A} = AD$ (Using sine law)

$$\Rightarrow PQ = (\sin A)(c \sin B) = \left(\frac{a}{2R} \right) (c) \left(\frac{b}{2R} \right)$$



$$= \frac{abc}{4R^2} = \frac{4R\Delta}{4R^2} = \frac{\Delta}{R}$$

7. Ans. (B)

Sol. $S_r = \frac{r^5 + 3r^4 + 2r^3 + 2}{r(r+1)(r+2)} = \frac{r^3(r+1)(r+2) + 2}{r(r+1)(r+2)}$

$$S_r = r^2 + \frac{2}{r(r+1)(r+2)}$$

$$\sum_{r=1}^{10} S_r = \sum_{r=1}^{10} r^2 + \sum_{r=1}^{10} \frac{2}{r(r+1)(r+2)}$$

$$= 385 + \sum_{r=1}^{10} \frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)}$$

$$= 385 + \frac{65}{132}$$

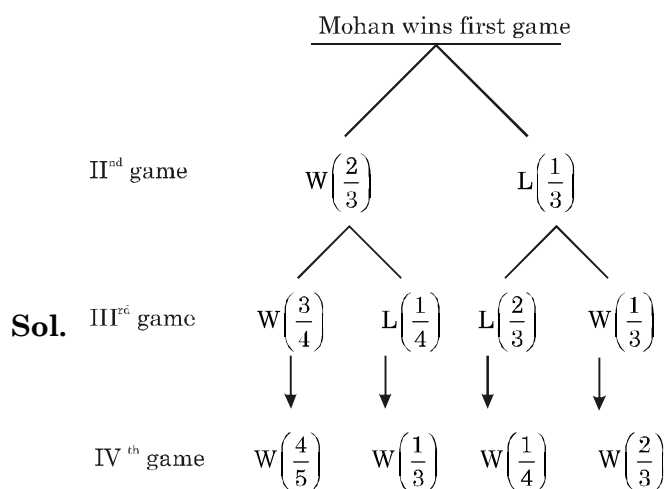
$$\left[\sum_{r=1}^{10} S_r \right] = 385$$

8. Ans. (A)

Sol. $V^2 = \frac{1}{36} [\vec{a} \vec{b} \vec{c}]^2 = \frac{1}{36} \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \frac{1}{12} \sqrt{3\sqrt{3}-5}$

Paragraph for Questions 9 and 10

9. Ans. (D)



$$\therefore \text{Prob (mohan win 4th game)} = \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} + \frac{2}{3} \times \frac{1}{4} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{3} \times \frac{1}{4} + \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} = \frac{79}{135}$$

10. Ans. (C)

Sol. Prob. (Mohan wins all 4 games when he

$$\text{won 4th game) = } \frac{\frac{2}{3} \times \frac{3}{4} \times \frac{4}{5}}{\frac{79}{135}} = \frac{54}{79}$$

Paragraph for Questions 11 and 12

11. Ans. (B, D)

Sol. Given system is

$$x + \lambda y + 3z = 15$$

$$3x + y - z = 3 + \lambda$$

$$4x + 3y + \lambda z = 17 + \mu$$

$$\text{Matrix of coefficients } A = \begin{bmatrix} 1 & \lambda & 3 \\ 3 & 1 & -1 \\ 4 & 3 & \lambda \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & \lambda & 3 \\ 3 & 1 & -1 \\ 4 & 3 & \lambda \end{vmatrix} = (\lambda + 3)(\lambda - 2)$$

Case - I

$$\lambda = 2 \Rightarrow |A| = 0$$

then, the given system will be

$$x + 2y + 3z = 15$$

$$3x + y - z = 5$$

$$4x + 3y + 2z = 17 + \mu$$

put $z = k$ in (1) and (2)

$$x + 2y = 15 - 3k$$

$$3x + y = 5 + k$$

$$\Rightarrow x = k - 1, y = 8 - 2k, z = k$$

substituting values of x, y and z in (3)

$$4x + 3y + 2z = 17 + \mu$$

$$4k - 4 + 24 - 6k + 2k = 17 + \mu$$

$$\mu = 3$$

$\Rightarrow$ System is consistent with infinite solutions, when $\lambda = 2$ and $\mu = 3$

Case - II

$$\lambda = -3 \Rightarrow |A| = 0$$

then, the given system will be

$$x - 3y + 3z = 15$$

$$3x + y - z = 0$$

$$4x + 3y - 3z = 17 + \mu$$

put $z = k$ in (1) and (2)

$$x - 3y = 15 - 3k$$

$$3x + y = k \Rightarrow x = \frac{3}{2}, y = k - \frac{9}{2}, z = k$$

substituting values of x, y and z in (3)

$$4x + 3y - 3z = 17 + \mu$$

$$6 + 3k - \frac{27}{2} - 3k = 17 + \mu \Rightarrow \mu = -49/2$$

$\Rightarrow$ System is consistent with infinite solutions, when $\lambda = -3$ and $\mu = -49/2$

Ans. of Q will be option (B) and (D)

12. Ans. (A, B, C, D)

Sol. The given system will have unique solution set, if

$$|A| \neq 0$$

$$(\lambda + 3)(\lambda - 2) \neq 0$$

$$\Rightarrow \lambda \neq -3 \text{ and } \lambda \neq 2$$

i.e. for $\lambda = -2$ or 3 or -1 or 4 system has unique solution set

Ans. (A, B, C, D)

Paragraph for Questions 13 and 14

13. Ans. (B)

14. Ans. (A)

Sol. Let $P(\alpha + 3, -2\alpha + 5, \alpha + 7)$ and

$Q(7\beta - 1, -6\beta - 1, \beta - 1)$ be the points on the given lines so that PQ is the line of shortest distance between the given lines

D.r's of $PQ = (\alpha - 7\beta + 4, -2\alpha + 6\beta + 6, \alpha - \beta + 8)$

Since PQ is perpendicular to the given lines

$$1(\alpha - 7\beta + 4) - 2(-2\alpha + 6\beta + 6) + 1(\alpha - \beta + 8) = 0$$

$$\Rightarrow 6\alpha - 20\beta = 0 \Rightarrow 3\alpha - 10\beta = 0 \quad \dots (1)$$

and,

$$\Rightarrow 7(\alpha - 7\beta + 4) - 6(-2\alpha + 6\beta + 6) + 1(\alpha - \beta + 8) = 0$$

$$\Rightarrow 7\alpha - 49\beta + 28 + 12\alpha - 36\beta - 36 + \alpha - \beta + 8 = 0$$

$$\Rightarrow 20\alpha - 86\beta = 0 \Rightarrow 10\alpha = 43\beta \quad \dots (2)$$

From (1) and (2) $\alpha = 0, \beta = 0$

$$P = (3, 5, 7), \quad Q = (-1, -1, -1)$$

Distance between PQ is $= 2\sqrt{29}$

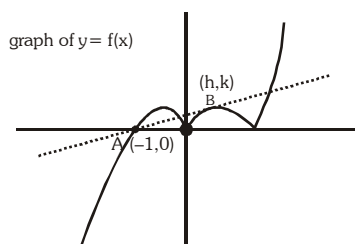
D.r's of $PQ = (-4, -6, -8) = (2, 3, 4)$

$$\text{Equation of PQ is } \frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$$

Paragraph for Questions 15 and 6

15. Ans. (C)

Sol.



$$\text{for (A) } f((x-2)^2 + 1) + f(-((x-2)^2 + 1)) = 0$$

$$f(t) + f(-t) = 0 \text{ where } t \geq 1, \text{ which is true}$$

from the graph.

for (B) obvious false

for (C) & (D)

$$y = p(x+1) \text{ passes through } (-1, 0)$$

$$m_{AB}$$

$$= \frac{k}{h+1} = 1 - 3h^2 \Rightarrow h - h^3 = (h+1)(1 - 3h^2)$$

$$\Rightarrow (h+1)(2h-1)(h+1) = 0$$

$$\Rightarrow h = \frac{1}{2} \Rightarrow (h, k) = \left(\frac{1}{2}, \frac{3}{8}\right)$$

$$\Rightarrow 0 < p < \frac{k}{h+1} \Rightarrow 0 < p < \frac{1}{4}$$

16. Ans. (B,D)

Sol. All the option can be verified from the graph

SECTION-II

1. Ans. 1.00

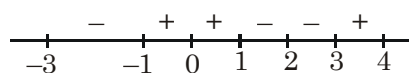
$$\text{Sol. } \alpha = \sum_{K=0}^{1006} \frac{(2013-K) - (K+1)}{(K+1) \binom{2013}{K+1}}$$

$$= \sum_{K=0}^{1006} \left(\frac{1}{\binom{2013}{K}} - \frac{1}{\binom{2013}{K+1}} \right)$$

$$\therefore \alpha = 1 - \frac{1}{\binom{2013}{1007}}$$

2. Ans. 4.00

$$\text{Sol. } f'(x) = \begin{cases} 4x(x^2 - 1) & ; -3 < x \leq 0 \\ \frac{2(1 - x^{4/3})}{x^{1/3}} & ; 0 < x < 2 \\ -1 & ; 2 < x < 3 \\ 1 & ; 3 < x < 4 \end{cases}$$



$\therefore$ extrema at $x = -1, 1, 2, 3$

3. Ans. 1.00

Sol. $f(x) = 3\tan x + x^3$

$$\Rightarrow f'(x) = 3\sec^2 x + 3x^2 > 0 \quad \forall x \in \left(0, \frac{\pi}{4}\right)$$

$$\therefore f(x) \uparrow \text{ on } x \in \left(0, \frac{\pi}{4}\right)$$

$$\text{Also, } f(0) = 0$$

$$\& f\left(\frac{\pi}{4}\right) > 2$$

So, by intermediate value theorem, $f(x) = 2$

will have exactly one solution in $\left(0, \frac{\pi}{4}\right)$

4. Ans. 6.00

Sol. Let $xy = \alpha$, $yz = \beta$, $zx = \gamma$

$$\therefore -5\alpha + 2\beta + \gamma = 2$$

$$2\alpha + \beta - \gamma = 1$$

$$6\alpha + \beta - 2\gamma = 3$$

So, on solving

$$\alpha = 2, \beta = 3, \gamma = 6$$

$$\therefore \alpha\beta\gamma = 36$$

$$\Rightarrow (xyz)^2 = 36$$

$$\Rightarrow xyz = \pm 6$$

$$\therefore |xyz| = 6$$

SECTION-IV

1. Ans. (A→Q ; B→Q,S ; C→Q,S ; D→S)

Sol. (A) For $y^2 = 4ax$

$$\text{Area} = 8a^2$$

$$\text{Here } a = 1$$

$$\therefore \text{Area} = 8 \text{ (sq. units)}$$

(B) The line $3x - y + 8 = 0$ passes through the focus

$(-2, 2)$ so the tangents at the end points on the chord intersect at right angles $\Rightarrow p = 8$

(C) $y^2 = k(x - 8/k)$ equation of directrix is

$$x - \frac{8}{k} = -\frac{k}{4} \Rightarrow x = \frac{8}{k} - \frac{k}{4}$$

$$\text{compare with } x = 1 \Rightarrow \frac{8}{k} - \frac{k}{4} = 1 \Rightarrow k = 4, -8$$

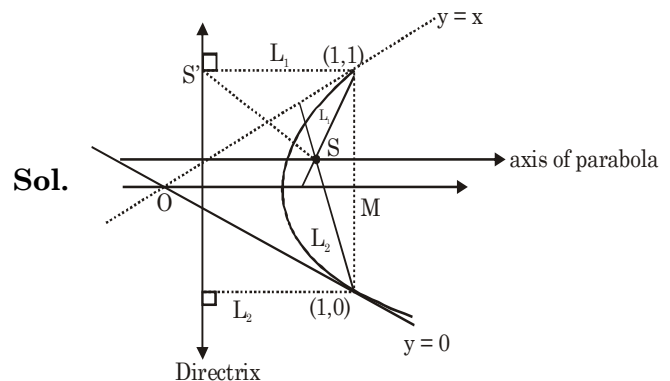
$$(D) PF_1 + PF_2 = 4$$

$$\text{Now, } \frac{PF_1 + PF_2}{2} \geq \sqrt{(PF_1)(PF_2)}$$

$$\Rightarrow 4 \geq (PF_1)(PF_2)$$

$$\therefore \boxed{(PF_1)(PF_2) \leq 4}$$

2. Ans. (A → P ; B → Q ; C → R ; D → S)



Sol.

In the figure; line MO is parallel to axis of parabola.

$\Rightarrow$ slope of axis $= \frac{1}{2}$. (Line joining intersection point of tangents and mid point of contact of tangents is parallel to axes of parabola)

L_1 & L_2 shown in figure are parallel to axis of parabola and image of L_1 and L_2 in corresponding tangents pass through focus.

$$L_1 \equiv x - 2y + 1 = 0$$

$$L_2 \equiv -2x + y + 1 = 0$$

$$L_2 \equiv x - 2y - 1 = 0$$

$$\& L_2 \equiv x + 2y - 1 = 0$$

$$S \equiv \left(\frac{3}{5}, \frac{1}{5}\right)$$

Now image of S lies on directrix

$$\Rightarrow S' \equiv \left(\frac{1}{5}, \frac{3}{5}\right)$$

$$\Rightarrow \text{Equation of directrix } 2x + y = 1$$

Equation of axis, passes through focus

$$S \equiv \left(\frac{3}{5}, \frac{1}{5}\right) \text{ and slope } \frac{1}{2} \text{ is } 10y = 5x - 1.$$

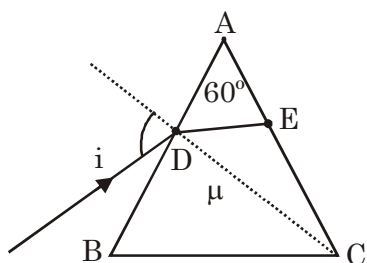
Solve equation of directrix and axis of parabola will get the foot of directrix.

now check options.

PAPER-2**PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A,B,C,D)****Sol.** For minimum distance of E from A, r, should be

$$\text{max possible} \Rightarrow r_1 = c_r = 45^\circ$$

$$\text{So } \angle ADE = 45^\circ$$



$$c_r = \text{critical Angle} = \sin^{-1}\left(\frac{1}{\mu}\right) = 45^\circ$$

$$\angle DAE = 60^\circ$$

$$AD = \frac{d}{2}$$

$$AE = ?$$

By sine rule

$$\frac{AE}{\sin 45^\circ} = \frac{AD}{\sin 75^\circ}$$

$$AE = \frac{d \sin 45^\circ}{2 \sin 75^\circ} = \frac{d}{2} \frac{\frac{1}{\sqrt{2}}}{\frac{\sqrt{3}+1}{2\sqrt{2}}}$$

$$= \frac{d}{2} (\sqrt{3} - 1)$$

$$(B) \mu = \frac{\sin\left(\frac{A + \delta_{\min}}{2}\right)}{\sin(A/2)} \Rightarrow \delta_{\min} = 30^\circ$$

$$\text{at } \mu = \sqrt{2}$$

(C) with $\mu = 2$, at 90° Angle of incidence, there is emergent ray present.2. **Ans. (B,C)****Sol.** Consider reduced mass

$$m = \frac{m_e \cdot m_c}{m_e + m_c} = \frac{m_e}{2}$$

$$\text{so replace } m_e \text{ by } \frac{m_c}{2}$$

$$\lambda_{\min} = \frac{2}{R}$$

3. **Ans. (A,C,D)****Sol.** Path of particle would be cycloidal, which is equivalent to ring rolling with speed

$$\left(\frac{E}{B}\right), \text{ so speed of particle is } 0 \leq v \leq \frac{2E}{B}.$$

4. **Ans. (A,B)****Sol.** Let potential of B and D is zero and common potential on capacitors is V, then at A and C it will be V.

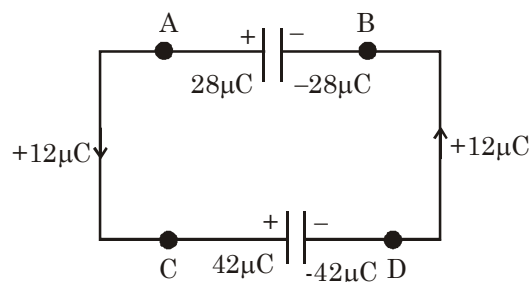
By charge conservation,

$$3V + 2V = 40 + 30$$

$$5V = 70 \Rightarrow V = 14 \text{ volt}$$

$$\text{Charge flow} = 40 - 28 = 12 \mu\text{C}$$

Now final charges on each plate is shown in the figure.



(ii) Heat produced

$$= \frac{1}{2} \times 2 \times (20)^2 + \frac{1}{2} \times 3 \times (10)^2 - \frac{1}{2} \times 5 \times (14)^2$$

$$= 400 + 150 - 490 = 550 - 490 = 60 \mu\text{J}$$

5. Ans. (A,B,C,D)

6. Ans. (C,D)

Sol. (A) at location of 500th maxima,

$$d \sin \theta = 500\lambda$$

$$\Rightarrow \sin \theta = \frac{500 \times 10^{-6}}{10^{-3}} = 0.5 \Rightarrow \theta = 30^\circ$$

$$\Rightarrow \frac{y}{D} = \tan 30^\circ \Rightarrow y = \frac{1}{2} \times \frac{1}{\sqrt{3}} = \frac{1}{2\sqrt{3}} \text{ m}$$

(B) & (C) Infront of one of the slit

$$\Delta x = \sqrt{D^2 + d^2} - D = \sqrt{100 + 21} - 10 = 1 \text{ cm}$$

$\Rightarrow \Delta x = \lambda$ so first maxima at location infront of slit.

(D) At location of 10th maxima

$$d \sin \theta = 10\lambda \Rightarrow \sin \theta = \frac{10 \times 10^{-6}}{10^{-3}} = 10^{-2}$$

$$\frac{y}{D} \approx \sin \theta \Rightarrow y = D \sin \theta = 0.5 \times 10^{-2} = 5 \text{ mm}$$

7. Ans. (A,B)

8. Ans. (A,B,D)

Sol. Torque about the point of contact is zero

$$\therefore mV_0 R - \frac{1}{2} mR^2 \omega_0 = mVR + \frac{1}{2} mR^2 \omega$$

$$\Rightarrow mV_0 R - \frac{1}{2} mR^2 \cdot \frac{3V_0}{R} = mVR + \frac{1}{2} mR^2 \cdot \frac{V}{R}$$

$$\Rightarrow -\frac{V_0}{2} = \frac{3V}{2}$$

$$\text{or } V = -\frac{V_0}{3} \text{ (when Rolling starts)}$$

If rolling starts at time t_1 and displacement s then.

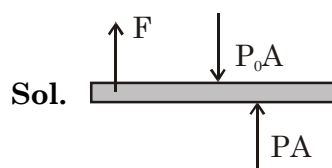
$$t_1 = \frac{-\frac{V_0}{3} - V_0}{(-\mu g)} = \frac{4V_0}{3\mu g}$$

$$\text{and } S = \frac{\left(V_0 - \frac{V_0}{3}\right)}{2} \times \frac{4V_0}{3\mu g} = \frac{4V_0^2}{9\mu g}$$

If t_2 is time taken to come to original position after rolling starts then

$$t_2 = \frac{\left(\frac{4V_0^2}{9\mu g}\right)}{\left(\frac{V_0}{3}\right)} = \frac{4V_0}{3\mu g}$$

9. Ans. (C)



Sol.

$$(P) \text{ As } V \uparrow P \downarrow \text{ So } F = (P_0 - P)A \uparrow$$

$$(Q) 2mv_{\text{avg}} \quad 2mv_{\text{avg}} \propto \sqrt{T} \text{ So constant}$$

$$(R) f \propto \frac{P}{v_{\text{avg}}} \Rightarrow P \downarrow \text{ So } F \downarrow \text{ (f = no. of collision per unit time)}$$

(S) Molar mass can't be predicted

(T) VT so $P \downarrow$

(U) As $V \uparrow$ so average distance $\uparrow$

10. Ans. (B)

Sol. (P) Can't be predicted

$$(Q) 2mv_{\text{avg}} \propto \sqrt{T} \text{ \& } Tv^{\gamma-1} = K \Rightarrow T \propto \frac{1}{v^{\gamma-1}}$$

So $V \uparrow$ & $T \downarrow$

$$(R) f \propto \frac{P}{v_{\text{avg}}} \propto \frac{1}{v^{\gamma} \cdot \sqrt{T}} \propto \frac{1}{v^{\frac{\gamma+1}{2}}}$$

So $v \uparrow$ $f \downarrow$

$$(S) \frac{dv}{dt} = 0 = \text{constant}$$

$$(T) P \propto \frac{1}{v^{\gamma}} \text{ So } P \downarrow$$

(U) $V \uparrow$ so average distance $\uparrow$

11. Ans. (C)

12. Ans. (D)

Sol. Q = B.E. of product – B.E. of reaction

Since Linear momentum is conserved,

$$\text{So, } P_B = P_C = P$$

$$kE_B + kE_C = Q$$

$$\frac{P^2}{2m_B} + \frac{P^2}{2m_C} = Q$$

$$\Rightarrow \frac{kE_B}{kE_C} = \frac{m_C}{m_B}$$

SECTION-II

1. Ans. 15.00

Sol. When R is not shunted

$$\frac{5}{R} = \frac{l_1}{100 - l_1} \quad \dots\dots(i)$$

When R is shunted with R (i.e.)

$$\frac{5}{(R/2)} = \frac{1.6l_1}{100 - 1.6l_1} \quad \dots\dots(ii)$$

Solve (i) and (ii) for R

2. Ans. 4.84

Sol. For resonance $\omega^2 = \frac{1}{LC}$

$$\text{i.e. } L_1 C_1 = L_2 C_2$$

$$\text{i.e. } L_2 = \left(\frac{C_1}{C_2} \right) L_1 = \left(\frac{C_1}{4C_1} \right) L_1 = \frac{L_1}{4}$$

$$\therefore |R| = |L|$$

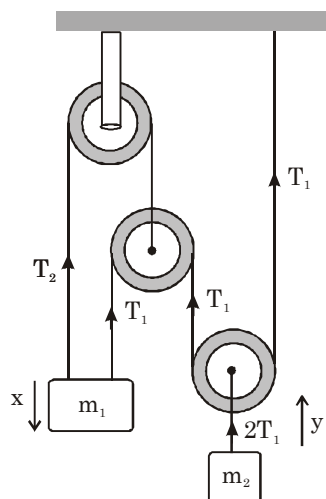
$$\therefore R_2 = \frac{R_1}{4}$$

$$\text{Resonant current } i = \frac{V}{R}$$

$$\therefore i_2 = 4i_1 = 4 \times 1.21 = 4.84 \text{ A}$$

3. Ans. 20.00

Sol.



$$T_2 = 2T_1$$

$$\text{Now, } m_1 g - T_1 - T_2 = m_1 a_1$$

$$m_1 g - 3T_1 = m_1 a_1 \quad \dots\dots(i)$$

$$2T_1 - m_2 g = m_2 a_2 \quad \dots\dots(ii)$$

$$\text{as } y = \left(\frac{3x}{2} \right)$$

$$\therefore 2a_2 = 3a_1 \quad \dots\dots(iii)$$

$$\text{By solving we get the answer } a_1 = \left(\frac{40}{19} \right)$$

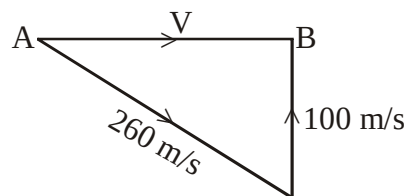
$$\text{m/sec}^2 \Rightarrow \frac{2n}{n-1} = \frac{40}{19}$$

$$\Rightarrow n = 20$$

4. Ans. 1.25

Sol. Speed of Sound in direction of AB

$$V = \sqrt{260^2 - 100^2} = 240 \text{ m/s}$$



$$\therefore t = \frac{300}{240} = 1.25$$

5. Ans. 1.50

Sol. $E_1 = -\frac{GMm}{2}$ ($r = 3R$)

$E_2 = -\frac{GMm}{2x}$ $x = \text{semi major axis}$

$\Delta E = E_2 - E_1 = \frac{GMm}{15R}$

6. Ans. 16.36

Sol. Let temperature of junction be $= \theta$

Heat flowing to junction = heat out flowing

$\frac{KA}{30}(30 - \theta) = \frac{KA}{20}(\theta - 20) + \frac{KA}{10}(\theta - 10)$

$\frac{(30 - \theta)}{3} = \frac{(\theta - 20)}{2} + \frac{(\theta - 10)}{1}$

$\frac{(30 - \theta)}{3} = \frac{\theta - 20 + 2\theta - 20}{2}$

$60 - 2\theta = 9\theta - 120$

$\frac{180}{11} = \theta$

$16.36^\circ\text{C} = \theta$

7. Ans. 0.10

Sol. $\tan \theta = \frac{r\omega^2}{g} = \frac{R(\sin \theta)\omega^2}{g}$

or $\cos \theta = \frac{g}{R\omega^2}$, i.e., $\frac{R-h}{R} = \frac{g}{R\omega^2}$

$h = R - (g/\omega^2)$

$g = \omega^2(R - h)$

$dg = -\omega^2 dh$, i.e., $(\Delta g)_{\min} = -(\omega^2)_{\min}(\Delta h)_{\min}$

So substituting the value of $\omega_{\min}$ and $(\Delta h)_{\min}$

$(\Delta g)_{\min} = -100 \times 10^{-3} \text{ m/s}^2$

8. Ans. 4.68

Sol. $w = \int PdV = \int_{V_0}^{2V_0} KV^{-2}dV$

$= -\left(\frac{K}{V}\right)_{V_0}^{2V_0} = \frac{K}{V_0} - \frac{K}{2V_0}$

$= \frac{P_0 V_0^2}{V_0} - \frac{P_0 V_0^2}{2V_0} = \frac{P_0 V_0}{2}$

$\therefore \frac{2.34}{x} = \frac{1}{2}$ or $x = 4.68$

9. Ans. 1.70 to 1.75

Sol. Natural frequency and period

$\omega_n = \sqrt{\frac{288}{22.08}} = 3.6116 \text{ rad/s}$

$\tau_n = \frac{2\pi}{\omega_n} = \frac{2\pi}{3.6116}$

$\tau_n = 1.740 \text{ s}$

10. Ans. 29.60

Sol. The interaction between the planet and the spacecraft is elastic, because the force of gravity is conservative. Thus kinetic energy is conserved in the interaction. Consider the problem a 1-dimensional collision, with A representing the spacecraft and B representing Saturn. Because the mass of Saturn is so much bigger than the mass of the spacecraft, Saturn's speed is not changed appreciably during the interaction. Use Eq., with $v_A = 10.4 \text{ km/s}$ and $v_B = v'_B = -9.6 \text{ km/s}$.

$v_A - v_B = -v'_A + v'_B \rightarrow v'_A = 2v_B - v_A$
 $= 2(-9.6 \text{ km/s}) - 10.4 \text{ km/s} = -29.6 \text{ km/s}$

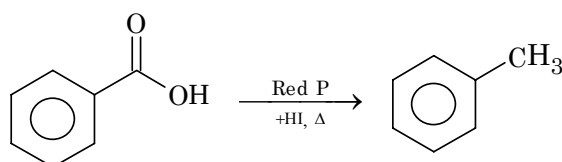
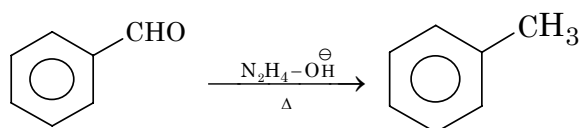
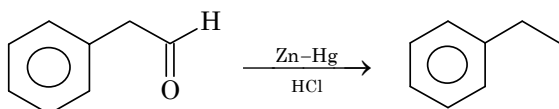
Thus there is almost a threefold increase in the spacecraft's speed.

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**

1. Ans. (B, C)
2. Ans. (A,B,D)
3. Ans. (A,B,C,D)

All statements are correct for starch.

4. Ans. (A,B,D)



5. Ans. (B,D)

Follow electronic configuration in each case.

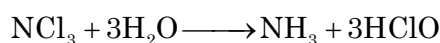
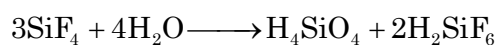
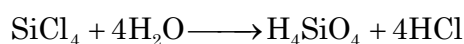
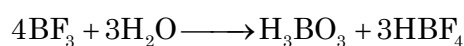
6. Ans. (A,B,C,D)

Dipole moment $\mu = e \times d$ and for polyatomic molecules, vector addition is applied.

7. Ans. (A,B,C,D)

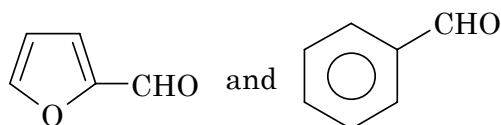
$$\text{At } 40 \text{ K, } \frac{\text{amount in vapour phase}}{\text{amount in liquid phase}} = \frac{bc}{ac} = \frac{2}{1}$$

8. Ans. (A,C,D)

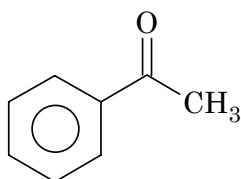


9. Ans. (A)

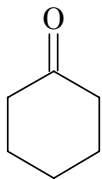
10. Ans. (B)



, Aromatic aldehyde can show Cannizzaro's reaction, Tollen's test, Perkins condensation and Benzoin condensation.



can undergo Aldol condensation and Iodoform test.



can undergo Aldol condensation.

11. **Ans. (B)**

12. **Ans. (D)**

Solution 11 & 12

11. Increase in temperature causes k_w to increase as dissociation of water is endothermic and the degree of ionisation also increases.

Pure water always stays neutral at any temperature but as ionization increases, $[H^+]$ increases and thus pH decreases.

Thus (I) matches with P, Q and S.

Dissociation of N_2O_4 into NO_2 is endothermic thus equilibrium constant and α increases with temperature. The reaction as written will go in forward direction with increase in T. Thus (II) matches with (Q) and (C) **Ans. (B)**.

12. H_2S is a polyprotic weak acid and $ka_1 \gg ka_2$. In such case concentration of second anion i.e. S^{2-} is equal to ka_2 , so $[S^{2-}] = 10^{-14}$ M. On dilution the pH will decrease for the solution of acid, but the solution will stay acidic. k_a value changes only with temperature.

The degree of ionisation increases with dilution for a weak electrolyte.

The direction of reaction can't be predicted as reaction is not written.

The correct match would be :

(III) $\rightarrow$ P, Q

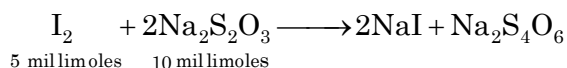
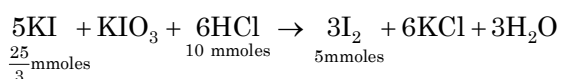
$NaHCO_3$ is amphoteric and thus pH is independent of concentration

$$pH = \frac{pk_a + pka_2}{2} = \frac{7 + 11}{2} = 9 \Rightarrow \text{basic}$$

Thus correct match is IV $\rightarrow$ U

SECTION-II

1. **Ans. (18.33)**



$$x = \frac{25}{3} \text{ and } y = 10 \Rightarrow x + y = \frac{25}{3} + 10 = \frac{55}{3} = 18.33$$

2. **Ans. (20.00)**

$$\text{For H : lines} = \frac{2(2-1)}{2} = 1$$

$$\text{For He}^+ : \text{lines} = \frac{4(4-1)}{2} = 6$$

$$\text{For Li}^{2+} : \text{lines} = \frac{6(6-1)}{2} = 15$$

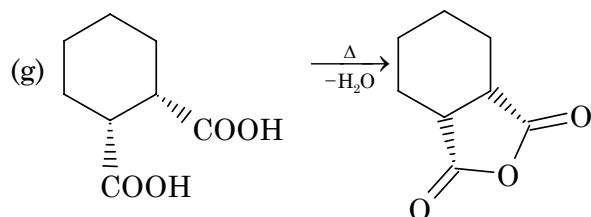
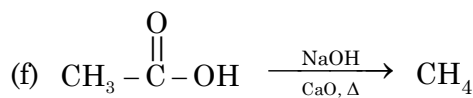
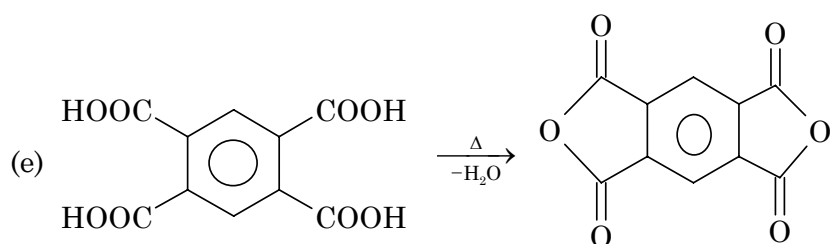
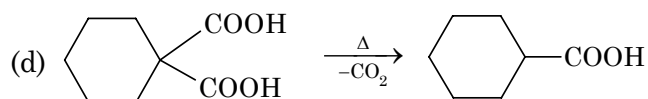
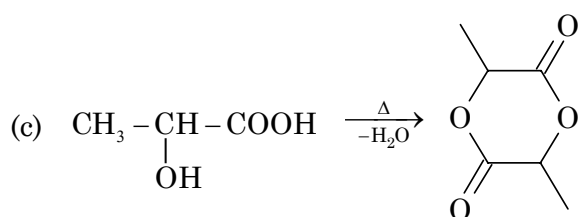
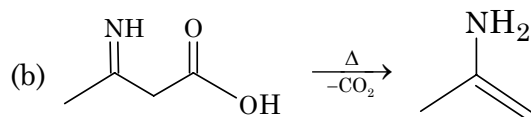
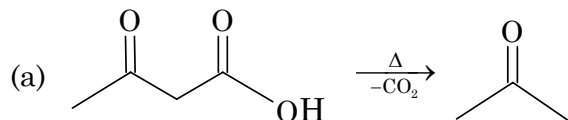
Total lines = 22

Common lines = 3

∴ different lines = 20.

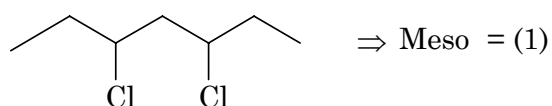
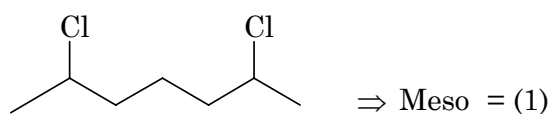
3. Ans. (4.00)

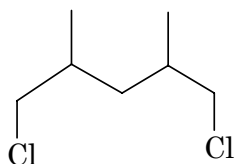
In reaction (a), (b), (d), (f) - decarboxylation occurs.



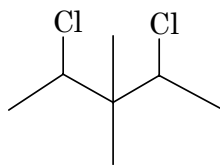
4. Ans. (5.00)

5. Ans. (6.00)

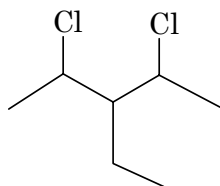




$\Rightarrow$ Meso = (1)



$\Rightarrow$ Meso = (1)



$\Rightarrow$ Meso = (2)

6. Ans. (3183.00)

7. Ans. (2.00)

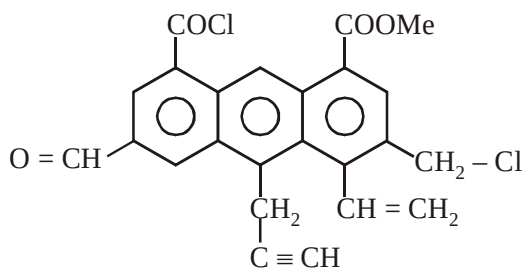
NO, N₂O

8. Ans. (3.00)

(K, Rb, Cs, C, Ba)

9. Ans.(4.00)

10. Ans.(7.00)



10. Ans.(7)

PART-3 : MATHEMATICS

SOLUTION

1. Ans. (B, C, D)

Sol. Let $g(x) = f(x) - 2x^2$

$\Rightarrow g(x) = 0$ has 3 real roots $x = 1, 2$ and 3

$\Rightarrow g'(x) = 0$ has atleast 2 real roots in $(1, 3)$

$\Rightarrow g''(x) = 0$ has atleast 1 real root in $(1, 3)$

$\Rightarrow f'(x) - 4 = 0$ has atleast 1 real root in $(1, 3)$ **Ans. (B)**

Now applying L.M.V.T. on $f(x)$ in $(2, 3)$

$\exists$ atleast one $x \in (2, 3)$ such that

$$f'(x) = \frac{f(3) - f(2)}{3 - 2} = \frac{18 - 8}{1} = 10$$

$\Rightarrow f'(x) = 10$ has atleast one root in $(2, 3)$

Ans. (C)

Again applying L.M.V.T. on $f(x)$ in $(1, 3)$

$\exists$ atleast one $x \in (1, 3)$ such that

$$f'(x) = \frac{f(3) - f(1)}{3 - 1} = \frac{18 - 2}{2} = 8$$

$\Rightarrow f'(x) = 8$ has atleast one root in $(1, 3)$

Ans. (D)

2. Ans. (A,C)

Sol. Any vector $\vec{r}$ in space can be written as

$$\vec{r} = \lambda \vec{a} + \mu \vec{b} + k(\vec{a} \times \vec{b})$$

take dot product with $\vec{a} \times \vec{b}$, we get $k = \frac{5}{4}$

$$\text{Let } \vec{r} \cdot \vec{a} = \lambda; \vec{r} \cdot \vec{b} = \mu$$

$$\therefore \vec{r} \cdot (3\vec{a} + 2\vec{b}) = 0 \Rightarrow 3\lambda + 2\mu = 0$$

$$\text{Given } \int_{-2\lambda}^{-\frac{4}{3}\mu} \frac{x+1}{x^2+1} dx = \int_{-2\lambda}^{2\lambda} \frac{x+1}{x^2+1} dx$$

$$= 2 \int_0^{2\lambda} \frac{dx}{x^2+1} = 2 \tan^{-1}(2\lambda) = \frac{\pi}{2}$$

$$\Rightarrow \lambda = \frac{1}{2} \text{ and } \mu = -\frac{3}{4}$$

$$\therefore \vec{r} = \frac{\vec{a}}{2} - \frac{3}{4}\vec{b} + \frac{5}{4}(\vec{a} \times \vec{b})$$

$$\text{and } |\vec{r}| = \sqrt{\frac{1}{4} + \frac{9}{16} + \frac{25}{16}} \Rightarrow |\vec{r}|^2 = \frac{19}{8}]$$

3. Ans. (B)

Sol. When $x < \frac{1}{2}$

$$(2x - 1)^2 - 3(1 - 2x) + 2 = 0$$

$$\Rightarrow 4x^2 - 4x + 1 - 3 + 6x + 2 = 0$$

$$\Rightarrow 4x^2 + 2x = 0$$

$$\Rightarrow 2x(2x + 1) = 0 \Rightarrow x = 0, -\frac{1}{2}$$

$$\text{when } x \geq \frac{1}{2} (2x - 1)^2 - 3(2x - 1) + 2 = 0$$

$$\Rightarrow 4x^2 - 4x + 1 - 6x + 3 + 2 = 0$$

$$\Rightarrow 4x^2 - 10x + 6 = 0$$

$$\Rightarrow 2x^2 - 5x + 3 = 0$$

$$\Rightarrow (2x - 3)(x - 1) = 0$$

$$\therefore x = 1, x = \frac{3}{2}$$

hence sum of the roots

$$= 0 - \frac{1}{2} + 1 + \frac{3}{2} = 2$$

(or assume equation as

$$|2x - 1|^2 - 3|2x - 1| + 2 = 0 \text{ \& put } |2x - 1| = y)$$

4. Ans. (A,B,C)

$$\text{Sol. } I = \int_{-\pi/4}^{n\pi-\pi/4} |\sin x + \cos x| dx = \sqrt{2} \int_{-\pi/4}^{n\pi-\pi/4} \left| \sin \left(x + \frac{\pi}{4} \right) \right| dx$$

$$\text{Put } x + \frac{\pi}{4} = t \quad dx = dt;$$

$$I = \sqrt{2} \int_0^{n\pi} |\sin t| dt = n\sqrt{2} \int_0^{\pi} |\sin t| dt$$

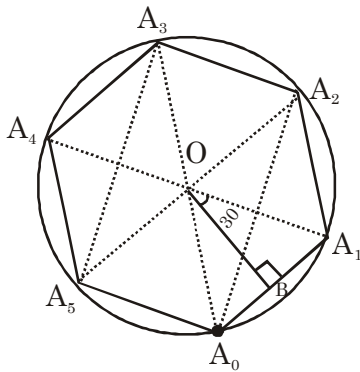
5. Ans. (A,B,C)

Sol. Z lies on the hyperbola with $(-1,0)$ and $(1,0)$

as focii hence least value of $|z|$ is $\frac{3}{4}$

6. Ans. (A, C, D)

Sol. $A_0A_1 = 2A_1B$



$$= 2(OA_1) \sin \frac{\pi}{6}$$

$$= 2 \cdot 1 \cdot \frac{1}{2}$$

$$= 1$$

Similarly

$$A_0A_2 = 2 \times 1 \times \sin\left(\frac{\pi}{3}\right) = \sqrt{3}$$

$$\text{and } A_0A_4 = A_0A_2 = \sqrt{3}$$

$$\text{Thus } A_0A_1 \times A_0A_2 \times A_0A_4 = 3 \quad \text{Ans. (A)}$$

Area of regular hexagon = $6 \times \text{Area of } \triangle A_0OA_1$

$$= 6 \times \frac{1}{2} \times A_0A_1 \times OB$$

$$= 6 \times \frac{1}{2} \times 1 \times (OA_1) \cos \frac{\pi}{6}$$

$$= 6 \times \frac{1}{2} \times 1 \times 1 \times \frac{\sqrt{3}}{2}$$

$$= \frac{3\sqrt{3}}{2}$$

$$\text{Area of quadrilateral } A_0A_2A_3A_5 = (A_0A_2)(A_2A_3)$$

$$= \sqrt{3} \times 1$$

$$= \sqrt{3}$$

Ans. (D)

$$\text{Let Area of } \triangle A_0A_1A_2 = \text{Area of } \triangle A_3A_4A_5 = \Delta$$

then $2\Delta + \text{Ar} \square A_0A_2A_3A_5 = \text{Area of hexagon}$

$$2\Delta + \sqrt{3} = \frac{3\sqrt{3}}{2}$$

$$\Rightarrow \Delta = \frac{\sqrt{3}}{4} \quad \text{Ans. (C)}$$

7. Ans. (D)

$$\text{Sol. } (AB^{-1}C)^{-1} = C^{-1}BA^{-1} = CBA$$

(for involutory matrix $C^{-1} = C$, $A^{-1} = A$)

8. Ans. (A, B, C)

$$\text{Sol. } p = \frac{3}{4} \quad q = \frac{1}{4} \quad n = 5$$

$$P(0) = {}^5C_0 \left(\frac{3}{4}\right)^0 \left(\frac{1}{4}\right)^5 = \frac{1}{1024}$$

$$P(1) = {}^5C_1 \left(\frac{3}{4}\right)^1 \left(\frac{1}{4}\right)^4 = \frac{15}{1024}$$

$$P(2) = {}^5C_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^3 = \frac{90}{1024} = \frac{45}{512}$$

$$P(3) + P(4) + P(5) = 1 - [P(0) + P(1) + P(2)]$$

$$= 1 - \left\{ \frac{1}{1024} + \frac{15}{1024} + \frac{90}{1024} \right\}$$

$$= 1 - \frac{106}{1024}$$

$$= \frac{918}{1024} = \frac{459}{512}$$

9. Ans. (B)

10. Ans. (A)

11. Ans. (D)

12. Ans. (A)

(I) Let the image of (2, 1, 4) in the plane

$2x - y + z + 5 = 0$ be (x_1, y_1, z_1) then

$$\frac{x_1 - 2}{2} = \frac{y_1 - 1}{-1} = \frac{z_1 - 4}{1} = \frac{-2\{2 \times 2 - 1 + 4 + 5\}}{(2)^2 + (-1)^2 + (1)^2} = -4$$

$$x_1 = -8 + 2 = -6; y_1 = 4 + 1 = 5; z_1 = -4 + 4 = 0$$

$\therefore$ The image is $(-6, 5, 0) = (\alpha_1, \alpha_2, \alpha_3)$

(II) Since the image point of (2, 1, 4) is

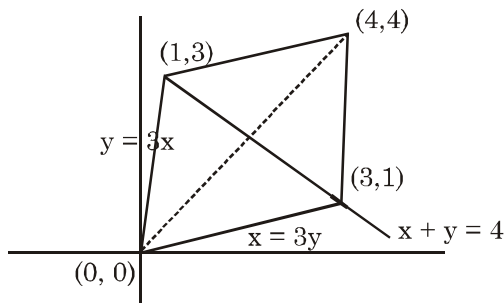
$(-6, 5, 0)$, the mid point of them is the foot of the perpendicular

$$\text{i.e., } (-2, 3, 2) = (\beta_1, \beta_2, \beta_3)$$

SECTION-II

1. Ans. (1.00)

Sol.



2. Ans. (2.00)

$$\text{Sol. } b = a + (n-1)d, c = \frac{n}{2}(a+b)$$

$$\therefore d = \frac{(a-b)(a+b)}{a+b-x\left(\frac{a+b}{2}\right)n} \Rightarrow d = \frac{2(a-b)}{2-nx}$$

$$\Rightarrow 2d - ndx = -2(n-1)d$$

$$\Rightarrow x = 2$$

3. Ans. (2.70)

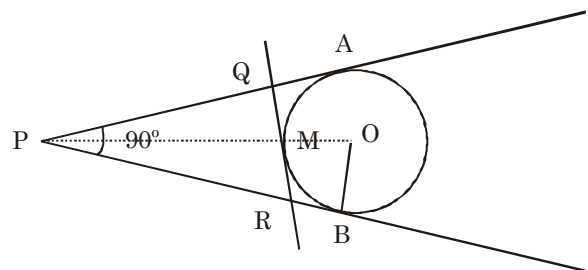
Sol. For non trivial solution.

$$\begin{vmatrix} a-1 & -1 & -1 \\ 1 & -(b-1) & 1 \\ 1 & 1 & -(c-1) \end{vmatrix} = 0 \Rightarrow ab + bc + ca = abc$$

$$\therefore \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1 \text{ using AM} \geq \text{GM} \quad abc \geq 27$$

4. Ans. (10.00)

Sol. Give circle will be ex-circle of ΔPQR



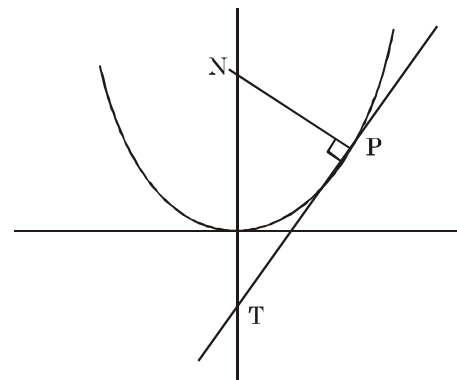
$$\text{so, } r_1 = s \tan \frac{A}{2}$$

$$s = \frac{r_1}{\tan \frac{A}{2}} = 5$$

$$\text{Perimeter} = 10$$

5. Ans. (0.80)

Sol.



$$\text{Using sine rule } \frac{PN}{\sin \angle PTN} = 2R$$

Circumcentre of triangle PTN will be at focus. So $R = 4$

$$\therefore \left(\frac{PN}{\sin(\angle PTN)} \right) = 8$$

6. Ans. (638.00)

$$\begin{aligned} \text{Sol. } \left(2 + \frac{x^4}{x^2+1} \right)^{10} &= \left(\frac{x^4 + 2x^2 + 2}{x^2+1} \right)^{10} = \left[(x^2+1) + \frac{1}{x^2+1} \right]^{10} \\ &\therefore \left((x^2+1) + \frac{1}{(x^2+1)} \right)^{10} = {}^{10}C_0 (x^2+1)^{10} + {}^{10}C_1 (x^2+1)^9 + {}^{10}C_2 (x^2+1)^8 + \dots + {}^{10}C_9 \frac{1}{(x^2+1)^9} + {}^{10}C_{10} \frac{1}{(x^2+1)^{10}} \\ \therefore A_0 &= {}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3 + {}^{10}C_4 + {}^{10}C_5 = 638 \end{aligned}$$

7. Ans. 7.00

Sol. For circle $\alpha^2 - \alpha = 2\alpha + 4$ ($a = b$)

$$(\alpha + 1)(\alpha - 4) = 0 \Rightarrow \alpha = -1, 4$$

$$\& \alpha^2 - 1 = 0 \Rightarrow \alpha = \pm 1 \text{ (h = 0)}$$

$$\therefore \alpha = -1$$

Equation becomes

$$2x^2 + 2y^2 - 12x + 8y + 4\lambda = 0$$

$$x^2 + y^2 - 6x + 4y + 2\lambda = 0$$

$$\text{Now } 9 < 9 + 4 - 2\lambda < 25, \therefore (r \in (3, 5))$$

$$-6 < \lambda < 2$$

8. Ans. (7.40)

$$\text{Sol. } f'(0^-) = 1 \& \lim_{x \rightarrow 0} \frac{1 - \cos\left(1 - \cos \frac{x}{2}\right)}{2^m x^n} = 1$$

$$\lim_{x \rightarrow 0} \frac{\left(1 - \cos \frac{x}{2}\right)^2}{2^m x^n} = 2$$

$\therefore$ for limit to exist $n = 4$.

$$\Rightarrow \frac{2}{2^{8+m}} = 1 \Rightarrow m = -7 \text{ and}$$

$$\text{hence } (n - 10m) = 74.$$

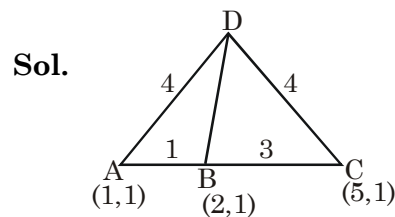
9. Ans. 2.00

$$\text{Sol. } x = \frac{\sqrt{7}}{2}$$

$$y = \frac{1}{8}$$

$$z = \frac{\sqrt{7}}{8}$$

10. Ans. 0.00



$$BD^2 = 4^2 + 1^2 - 2(1)(4)\cos 60^\circ$$

$$\Rightarrow BD = \sqrt{3}$$

$$\text{ar}(\triangle ABD) = \frac{1}{2}(4)(1)\sin 60^\circ = \sqrt{3}$$

$$\text{ar}(\triangle BCD) = \frac{1}{2}(4)(3)\sin 60^\circ = 3\sqrt{3}$$

$$\therefore r_1 r_2 = \frac{\sqrt{3}}{\left(\frac{4+1+\sqrt{13}}{2}\right)} \frac{3\sqrt{3}}{\left(\frac{4+3+\sqrt{13}}{2}\right)}$$

$$= \frac{36}{(5+\sqrt{13})(7+\sqrt{13})} = \frac{36}{48+12\sqrt{13}}$$

$$= \frac{3}{(4+\sqrt{13})} \times \frac{(4-\sqrt{13})}{(4-\sqrt{13})} = (4-\sqrt{13})$$